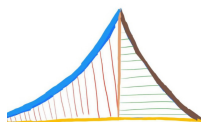


Matrices In Geometry

1



G. V. V. Sharma

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE AG 2025

A. Formulae

1.1.1 Z transform:

$$M_{Y_1}(Z) = E(Z^{-Y_1}) = \sum P_{Y_1}(k) Z^{-k} \quad (1.1.1.1)$$

If $Y_1, Y_2, \dots, Y_n$ are independent events:

$$M_Y(Z) = M_{Y_1}(Z) M_{Y_2}(Z) \dots M_{Y_n}(Z) \quad (1.1.1.2)$$

Laplace transforms of some common functions:

1.1.2

$$\cos(at) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2} \quad (1.1.2.1)$$

Proof.

$$\cos(at) \xleftrightarrow{\mathcal{L}} \int_0^\infty \cos(at) e^{-st} dt = I \quad (1.1.2.2)$$

$$I = \int_0^\infty \frac{d(\sin(at))}{a} e^{-st} \quad (1.1.2.3)$$

$$= \frac{\sin(at)}{a} e^{-st} \Big|_0^\infty - \frac{1}{a} \int_0^\infty \sin(at) e^{-st} (-s) dt \quad (1.1.2.4)$$

$$= 0 + \frac{s}{a} \int_0^\infty \frac{d(-\cos(at))}{a} e^{-st} \quad (1.1.2.5)$$

$$= -\left(\frac{s}{a^2}\right) \left\{ \cos(at) e^{-st} \Big|_0^\infty - \int_0^\infty \cos(at) e^{-st} (-s) dt \right\} \quad (1.1.2.6)$$

$$= -\left(\frac{s}{a^2}\right) \{(0 - 1) - (-s)I\} \quad (1.1.2.7)$$

$$= -\left(\frac{s}{a^2}\right) \{-1 + sI\} \quad (1.1.2.8)$$

$$I = \frac{s}{a^2} - \frac{s^2}{a^2} (I) \quad (1.1.2.9)$$

$$\left(\frac{a^2 + s^2}{a^2}\right) I = \frac{s}{a^2} \quad (1.1.2.10)$$

$$I = \frac{s}{s^2 + a^2} \quad (1.1.2.11)$$

$$\sin(at) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \quad (1.1.2.12)$$

Proof.

$$\sin(at) \xleftrightarrow{\mathcal{L}} \int_0^\infty \sin(at) e^{-st} dt = I \quad (1.1.2.13)$$

$$I = \int_0^\infty \frac{d(-\cos(at))}{a} e^{-st} \quad (1.1.2.14)$$

$$= \frac{-\cos(at)}{a} e^{-st} \Big|_0^\infty - \frac{1}{a} \int_0^\infty -\cos(at) e^{-st} (-s) dt \quad (1.1.2.15)$$

$$= \frac{1}{a} - \frac{s}{a} \int_0^\infty \frac{d(\sin(at))}{a} e^{-st} \quad (1.1.2.16)$$

$$= \frac{1}{a} - \left(\frac{s}{a^2}\right) \left\{ \sin(at) e^{-st} \Big|_0^\infty - \int_0^\infty \sin(at) e^{-st} (-s) \right\} \quad (1.1.2.17)$$

$$= -\left(\frac{s}{a^2}\right) \{(0 - 0) - (-s) I\} \quad (1.1.2.18)$$

$$= -\left(\frac{s}{a^2}\right) \{0 + sI\} \quad (1.1.2.19)$$

$$I = \frac{a}{a^2} - \frac{s^2}{a^2} (I) \quad (1.1.2.20)$$

$$\left(\frac{a^2 + s^2}{a^2}\right) I = \frac{a}{a^2} \quad (1.1.2.21)$$

$$I = \frac{a}{s^2 + a^2} \quad (1.1.2.22)$$

$$\sinh(at) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 - a^2} \quad (1.1.2.23)$$

Proof.

$$\sinh(at) \xleftrightarrow{\mathcal{L}} \int_0^\infty \frac{e^{at} - e^{-at}}{2} e^{-st} dt \quad (1.1.2.24)$$

$$= \frac{1}{2} \int_0^\infty e^{(a-s)(t)} dt - \frac{1}{2} \int_0^\infty e^{(-a-s)(t)} dt \quad (1.1.2.25)$$

$$= \frac{1}{2} \left\{ \frac{e^{(a-s)(t)}}{a-s} \right\} \Big|_0^\infty - \frac{1}{2} \left\{ \frac{e^{(-a-s)(t)}}{-a-s} \right\} \Big|_0^\infty$$

(assume $re(s) > re(a)$ for integral to be finite)

$$= \frac{1}{2} \left(\frac{1}{s-a} - \frac{1}{s+a} \right) \quad (1.1.2.26)$$

$$= \frac{1}{2} \left(\frac{2a}{s^2 - a^2} \right) \quad (1.1.2.27)$$

$$I = \frac{a}{s^2 - a^2} \quad (1.1.2.28)$$

$$\cosh (at) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 - a^2} \quad (1.1.2.29)$$

Proof.

$$\cosh (at) \xleftrightarrow{\mathcal{L}} \int_0^\infty \frac{e^{at} + e^{-at}}{2} e^{-st} dt \quad (1.1.2.30)$$

$$= \frac{1}{2} \int_0^\infty e^{(a-s)(t)} dt + \frac{1}{2} \int_0^\infty e^{(-a-s)(t)} dt \quad (1.1.2.31)$$

$$= \frac{1}{2} \left\{ \frac{e^{(a-s)(t)}}{a-s} \right\} \Big|_0^\infty + \frac{1}{2} \left\{ \frac{e^{(-a-s)(t)}}{-a-s} \right\} \Big|_0^\infty$$

(assume $re(s) > re(a)$ for integral to be finite)

$$= \frac{1}{2} \left(\frac{1}{s-a} + \frac{1}{s+a} \right) \quad (1.1.2.32)$$

$$= \frac{1}{2} \left(\frac{2s}{s^2 - a^2} \right) \quad (1.1.2.33)$$

$$I = \frac{s}{s^2 - a^2} \quad (1.1.2.34)$$

Sum of n terms of a gp

$$\sum_{i=1}^n q^i = q \frac{q^n - 1}{q - 1} \quad (1.1.2.35)$$

Expansions:

$$\sin (x) = \sum_{i=1}^n -1^{i+1} \frac{x^{2i-1}}{(2i-1)!} \quad (1.1.2.36)$$

$$\cos (x) = \sum_{i=1}^n -1^{i+1} \frac{x^{i-1}}{i-1!} \quad (1.1.2.37)$$

Euler's Theorem for homogeneous functions:

It states that if a function $f(x, y)$ is homogeneous of degree n , then the following condition holds:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n f \quad (1.1.2.38)$$

$$f'(x) \xleftrightarrow{\mathcal{L}} sG(s) - f(0) \quad (1.1.2.39)$$

Proof. Let $G(s)$ be the Laplace transform of $y=f(x)$:

$$y \xleftrightarrow{\mathcal{L}} G(s) \quad (1.1.2.40)$$

By definition of Laplace transform,

$$G(s) = \int_0^{\infty} f(x) e^{-sx} dx \quad (1.1.2.41)$$

integrating by parts,

$$G(s) = \int_0^{\infty} f(x) d\left(\frac{e^{-sx}}{-s}\right) \quad (1.1.2.42)$$

$$= f(x) \frac{e^{-sx}}{-s} \Big|_0^{\infty} + \int_0^{\infty} f'(x) \frac{e^{-sx}}{s} \quad (1.1.2.43)$$

$$= \left(\frac{f(\infty) e^{-\infty} - f(0)}{-s} \right) + \frac{1}{s} \mathcal{L}\{f'(x)\} \quad (1.1.2.44)$$

$$sG(s) = f(0) + \mathcal{L}\{f'(x)\} \quad (1.1.2.45)$$

$$f'(x) \xleftrightarrow{\mathcal{L}} sG(s) - f(0) \quad (1.1.2.46)$$

The Laplace transform is a linear function

The inverse Laplace transform is a linear function

Laplace transform of e^{ax}

$$e^{ax} \xleftrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.1.2.47)$$

Proof.

$$e^{ax} \xleftrightarrow{\mathcal{L}} \int_0^{\infty} e^{at} e^{-st} dt \quad (1.1.2.48)$$

$$= \int_0^{\infty} e^{(a-s)(t)} dt \quad (1.1.2.49)$$

$$= \frac{1}{a-s} (e^{(a-s)(t)} \Big|_0^{\infty}) \quad (1.1.2.50)$$

$$= \frac{1}{a-s} (0 - 1) \quad (\text{Assuming that } \operatorname{re}(s) > \operatorname{re}(a))$$

$$= \frac{1}{s-a} \quad (1.1.2.51)$$

Euler's forward difference method: The output of $f(x)$ at $x = x_0$ is y_0 , then,

$$y_1 = y_0 + hf'(x_0, y_0)$$

where y_1 is the value of the function at $x = x_0 + h$

Newton Raphson method: The improved guess of the root of $f(x)$ is given by:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Where x_0 is the old guess and x_1 is the improved guess.

B. Solved Questions

1.2.1 A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once? (GATE AG 2025)

- a) $\frac{75}{216}$
- b) $\frac{1}{6}$
- c) $\frac{18}{216}$
- d) $\frac{25}{216}$

Solution: The sample space is $S \in \{1, 2, 3, 4, 5, 6\}$. Let us define a variable Y_1 as follows:

$$Y_1 = \begin{cases} 1 & X_1 = 6 \\ 0 & X_1 \neq 6 \end{cases} \quad (1.2.1.1)$$

Now, The PMF is defined as:

$$P_{Y_1}(k) = \begin{cases} \frac{1}{6} & k = 1 \\ \frac{5}{6} & k = 0 \end{cases} \quad (1.2.1.2)$$

$$(1.2.1.3)$$

$$\text{Let } p, q \text{ be } \frac{1}{6} \text{ and } \frac{5}{6} \text{ respectively.} \quad (1.2.1.4)$$

Let us define Y_2 and X_3 as the random variables for the 2nd and 3rd dice roll, respectively. Let Y denote the random variable for the combined triple dice roll. We write

$$Y = Y_1 + Y_2 + Y_3 \quad (1.2.1.5)$$

Writing the Z-Transform (see (1.1.1)) for Y

$$M_{Y_1}(Z) = (P_{Y_1}(1)Z^{-1} + P_{Y_1}(0)Z^{-0}) \quad (1.2.1.6)$$

$$= Z^{-1} + q \quad (1.2.1.7)$$

We know that Y_1, Y_2, Y_3 are independent events. From (1.1.1.2)

$$M_Y(Z) = M_{Y_1}(Z) M_{Y_2}(Z) M_{Y_3}(Z) \quad (1.2.1.8)$$

$$M_Y(Z) = (pZ^{-1} + q)^3 \quad (1.2.1.9)$$

$$= \sum_{k=0}^3 {}^3C_k q^{3-k} (pZ^{-1})^k \quad (1.2.1.10)$$

From (1.1.1.1)

$$= \sum P_Y(k) Z^{-k} \quad (1.2.1.11)$$

So, we can say:

$$P_Y(k) = {}^3C_k q^{3-k} p^k \quad (1.2.1.12)$$

The above situation is called as a binomial distribution. We need the situation where

p appears exactly once. So, we consider only the term where the power of p is 1. The required term is : ${}^3C_1 p q^2$. Substituting the values of p and q , we get:

$${}^3C_1 \frac{1}{6} \left(\frac{5}{6}\right)^2 = \frac{75}{216} \quad (1.2.1.13)$$

Therefore, the required answer is $\frac{75}{216}$. The following code simulates the result :

codes/Probability.c

1.2.2 If a, b, and c are the roots of $2x^3 - 3x^2 + px - 1 = 0$ and sum of the two roots is 1, the value of p is: (GATE AG 2025)

- a) 2
- b) 1
- c) 3
- d) $\frac{1}{3}$

Solution:

- a) Let the roots whose sum is 1 be a and b . Then, by using the properties of coefficients of an equation, we write:

$$-\left(\frac{-3}{2}\right) = a + b + c \quad (1.2.2.1)$$

$$\frac{3}{2} = 1 + c \quad (1.2.2.2)$$

$$c = \frac{1}{2} \quad (1.2.2.3)$$

And

$$abc = -\left(\frac{-1}{2}\right) \quad (1.2.2.4)$$

$$ab \frac{1}{2} = \frac{1}{2} \quad (1.2.2.5)$$

From (1.2.2.3),

$$ab = 1 \quad (1.2.2.6)$$

Also,

$$ab + bc + ac = \left(\frac{p}{2}\right) \quad (1.2.2.7)$$

From (1.2.2.6) and (1.2.2.3), we write

$$1 + c(a + b) = \frac{p}{2} \quad (1.2.2.8)$$

$$1 + \frac{1}{2}(1) = \frac{p}{2} \Rightarrow 3 = p \quad (1.2.2.9)$$

Therefore, the value of p is 3.

- b) Calculating the roots by newton-raphson method (see (1.1.2)). The given cubic

equation is $:2x^3 - 3x^2 + 3x - 1$. Therefore, the difference equation becomes:

$$x_{n+1} = x_n + \frac{2x^3 - 3x^2 + 3x - 1}{6x^2 - 6x + 3} \quad (1.2.2.10)$$

With an initial guess of 4, running this difference equation 6 times gives us the real root, which is $\frac{1}{2}$

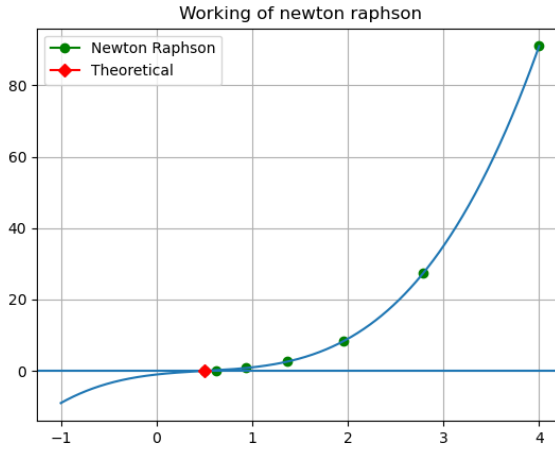


Fig. 1.2.2.1. The depiction of newton raphson method guessing the root

The following code verifies the result:

```
codes/raph.py
```

1.2.3 If the matrix $\begin{pmatrix} x & -3 \\ 2 & x-5 \end{pmatrix}$ is singular, sum of the values of x is: (GATE AG 2025)

- a) 5
- b) 6
- c) 0
- d) -1

Solution: A matrix is said to be singular if its row echelon form has a row full of zeroes. Finding row echelon form of the given matrix:

$$\begin{pmatrix} x & -3 \\ 2 & x-5 \end{pmatrix} \xrightarrow{R_2 \rightarrow \frac{R_2}{2}} \begin{pmatrix} x & -3 \\ 1 & \frac{(x-5)}{2} \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 - xR_2} \begin{pmatrix} x - x \cdot 1 & \frac{-6 - (x-5)}{2} \\ 1 & \frac{(x-5)}{2} \end{pmatrix} \xrightarrow{R_2 \rightarrow R_1 R_2 \rightarrow R_1} \begin{pmatrix} 1 & \frac{(x-5)}{2} \\ 0 & \frac{(-6 - x^2 + 5x)}{2} \end{pmatrix}$$

Equating $\frac{(-6-x^2+5x)}{2}$ to zero gives us:

$$-6 - x^2 + 5x = 0 \quad (1.2.3.1)$$

$$x^2 - 5x + 6 = 0 \quad (1.2.3.2)$$

So, the sum of values of x is 5. The following code converts a numerical matrix into its row echelon form. So, we can verify that the values of $(x = 2, 3)$ give us a matrix with a row of zeroes. The following code verifies the result:

```
codes/echelon.py
```

1.2.4 $\frac{s}{s^2+a^2}$ is the Laplace transform of: (GATE AG 2025)

- a) $\cos(at)$
- b) $\sin(at)$
- c) $\sinh(at)$
- d) $\cosh(at)$

Solution: From (1.1.1),

$$\cos(at) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2} \quad (1.2.4.1)$$

1.2.5 The median of the following set of numbers: 154, 130, 144, 137, 156, 146, 138, 149, 160, 138 is: (GATE AG 2025)

- (A) 145.0
- (B) 145.2
- (C) 151.0
- (D) 146.0

Solution: To find the median, we need to arrange the data in ascending or descending order. After arranging them in ascending order, the data is as follows

$$130, 137, 138, 138, 144, 146, 149, 154, 156, 160 \quad (1.2.5.1)$$

The number of elements in the data set are 10. Since it is an even number, the median (M) is defined as the average of the middle 2 terms (in this case, average of 5th and 6th terms). Therefore, the median(M) is $\frac{144+146}{2} = 145.0$. The following codes verify the result:

```
codes/Sorter.py
codes/Quicksort.c
```

1.2.6 The series $\sum_{n=0}^r q^n = 1 + q + q^2 + \dots$ has the sum: (GATE AG 2025)

- a) $\frac{1}{1-q} - \frac{q^{r+1}}{1-q}$
- b) $\frac{1}{q-1} + \frac{q^{r+1}}{q-1}$
- c) $\frac{1}{1-q} + \frac{q^{r+1}}{1-q}$
- d) $\frac{1}{q-1}$

Solution: The given sequence is a geometric progression with initial term as 1 and common ratio as q and the number of terms are $r + 1$. (The series converges only if

$q < 1$). Applying formula (1.1.2.35), we get :

$$1 \frac{q^{r+1} - 1}{q - 1} \quad (1.2.6.1)$$

$$= \frac{1 - q^{r+1}}{1 - q} \quad (1.2.6.2)$$

$$= \frac{1}{1 - q} - \frac{q^{r+1}}{1 - q} \quad (1.2.6.3)$$

Therefore, the answer is option A. The following code verifies the result:

codes/gp.py

1.2.7 The value of $\lim_{x \rightarrow 0} \frac{x \cos(x) - \sin(x)}{x^2 \sin(x)}$ (GATE AG 2025)

- a) $-\frac{1}{3}$
- b) $\frac{1}{3}$
- c) 3
- d) -3

Solution: We use the idea of taylor series expansion (converges if $x < 1$) to solve the problem.

$$\frac{x (\cos(x)) - \sin(x)}{x^2 \sin(x)} \quad (1.2.7.1)$$

$$(1.2.7.2)$$

From (1.1.2)

$$= \frac{x(1 - \frac{x^2}{2!} \dots) - (x - \frac{x^3}{3!} \dots)}{x^2(x - \frac{x^3}{3!} \dots)} \quad (1.2.7.3)$$

When x approaches zero, we can ignore the higher power terms.

$$= \frac{(x - \frac{x^3}{2}) - (x - \frac{x^3}{6})}{x^3(1 - \frac{x^2}{6})} \quad (1.2.7.4)$$

$$= \frac{-\frac{1}{3}}{1 - \frac{x^2}{6}} \quad (1.2.7.5)$$

Substituting x as zero gives us the limit as $-\frac{1}{3}$.

The following code simulates the result:

codes/Plottinglimit.py

The output graph is shown below:

1.2.8 If $u = x^4 + y^4 + 3x^2y^2$, the value of $\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}\right)^{-1}$ (GATE AG 2025)

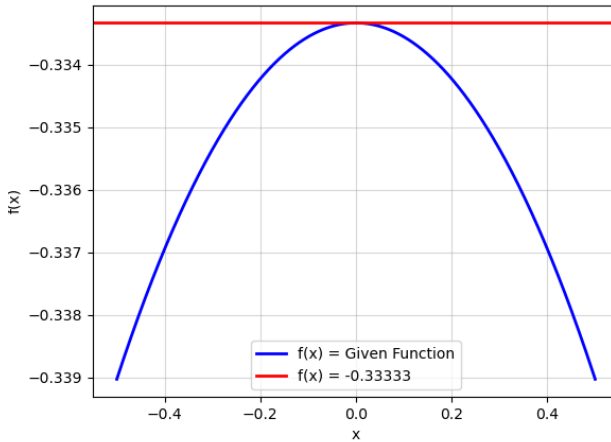


Fig. 1.2.7.1. The plot depicting the approaching of the limit to $-\frac{1}{3}$

- a) $4u$
- b) 0
- c) $\frac{1}{4u}$
- d) $2u$

Solution: We use Euler's theorem for homogeneous functions.(Refer to (1.1.2.38) for the theorem)

$$u = g(x, y) = x^4 + y^4 + 3x^2y^2 \quad (1.2.8.1)$$

$$g(\alpha x, \alpha y) = \alpha^4 (x^4 + y^4 + 3x^2y^2) \quad (1.2.8.2)$$

Therefore, u is a homogeneous function of degree 4. Since we know that the given equation is homogeneous and of degree 4, we can apply this theorem.

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 4u \quad (\text{Refer equation (1.1.2.38)})$$

Since they asked for its reciprocal, the answer is $\frac{1}{4u}$

1.2.9 The value of y at $x = 0.3$ and $y(x = 0) = 0$ for the differential equation $\frac{dy}{dx} = 3y + 2e^x$ is _____. (Rounded off to 3 decimal places) (GATE AG 2025)

Solution:

a) Let us define a function $G(s)$ such that:

$$y \xleftrightarrow{\mathcal{L}} G(s) \quad (1.2.9.1)$$

$$y' \xleftrightarrow{\mathcal{L}} sG(s) \quad (\text{Refer (1.1.2.39)})$$

Applying the Laplace transform:

$$sG(s) = 3G(s) + 2\mathcal{L}\{e^x\} \quad (\text{From (1.1.2)})$$

Now, we evaluate $G(s)$

$$G(s)(s-3) = \frac{2}{s-1} \quad (1.2.9.2)$$

$$G(s) = \frac{2}{(s-1)(s-3)} \quad (1.2.9.3)$$

Using partial fractions:

$$G(s) = \left(\frac{1}{s-3} - \frac{1}{s-1} \right) \quad (1.2.9.4)$$

Finding y using inverse Laplace transform:

$$y \xleftrightarrow{\mathcal{L}} G(s) \quad (1.2.9.5)$$

From (1.1.2) and (1.1.2.47), we can write:

$$y = e^{3x} - e^x \quad (1.2.9.6)$$

finding y based on initial conditions:

$$x = 0.3 \quad (\text{given})$$

$$y = e^{3 \times 0.3} - e^{0.3} \quad (1.2.9.7)$$

$$= 1.110 \quad (1.2.9.8)$$

- b) Using forward euler method (see (1.1.2)) to find the graph. The difference function of forward euler becomes:

$$y_{n+1} = y_n + h(3y_n + 2e^x) \quad (1.2.9.9)$$

Using it, the following graph is obtained:

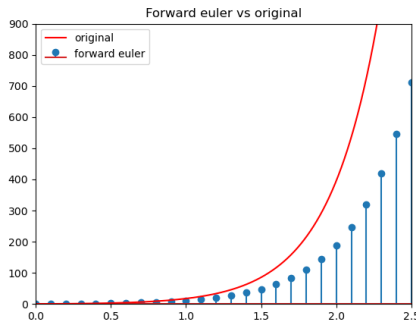


Fig. 1.2.9.1. The plot depicting the divergence of forward euler from original at large values

The following code simulates the result:

```
codes/diffeq.py
```